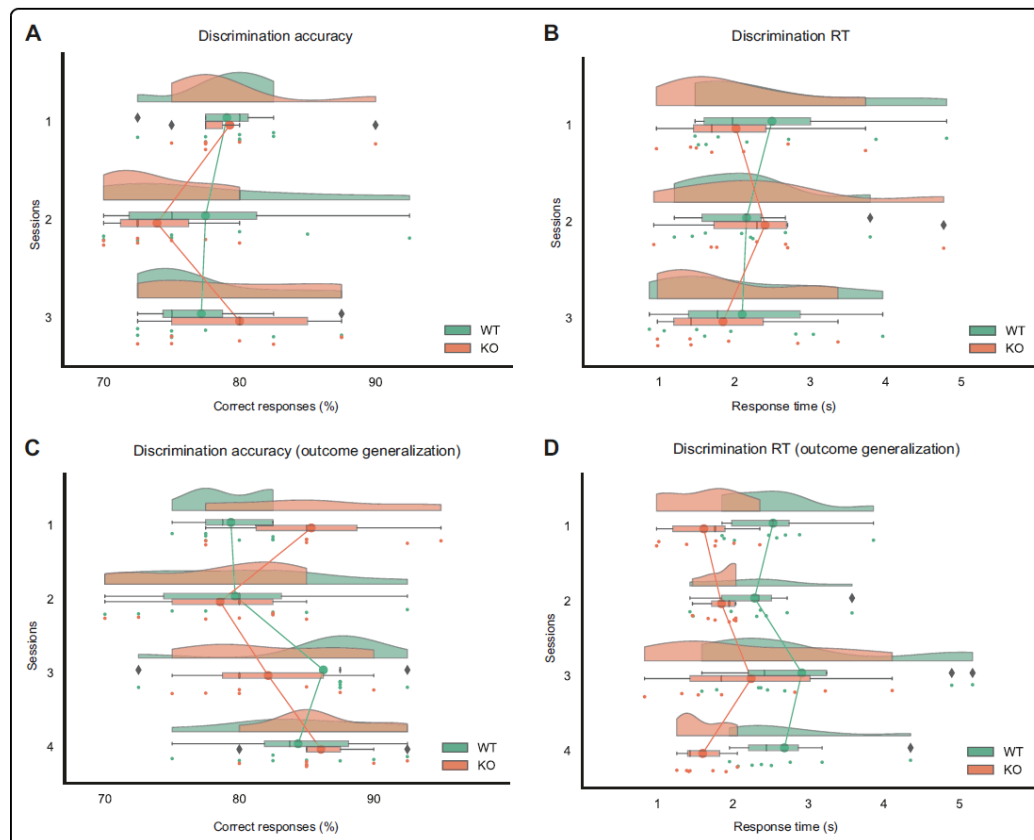


Article Critique #1

The researchers of this study aimed to investigate 5-HTT's involvement in psychiatric disorders (anxiety, depression, drug abuse, etc.). Previous research had shown that knocking-out the 5-HTT gene in rats (causing inhibition serotonin transporter) resulted in the expression of depression/anxiety-like symptoms as a result of the mutagenesis they underwent. Additionally, some other articles provided evidence for 5-HTT's involvement in reward learning. Together these implied to the researchers that 5-HTT would be a promising area of focus for this study. The other primary motivation for this research paper was to address whether or not it would be appropriate to make "generalization" a new dimension for the RDoC framework (since inability to generalize is considered "maladaptive" and is proposed to a cause for mental disorders in both humans and animals). If the data Guo and his team collected proved the necessity of 5-HTT in generalization/learning, this could be revolutionary in providing new insight as to how we diagnose these disorders, but also in how we treat them

The researchers wanted to see exactly how knocking out the 5-HTT gene in rats would change their outcome and stimulus generalization. They used 20 male rats—half were "wild-type" (WT) and half were 5-HTT "knockout" (KO). The rats were trained on a touchscreen task using Gabor patches, which are visual patterns often used in human studies. The experiment moved through several phases: first, basic learning (instrumental), then telling two stimuli apart (discrimination), then seeing how they handled a drop in reward predictability (outcome generalization), and finally, how they reacted to brand new, ambiguous patterns (stimulus generalization). The results showed that while both groups of rats were equally good at learning the initial tasks, they handled environmental changes differently. Specifically, in the outcome generalization phase, the WT rats adapted over time, while the KO rats were faster at first but failed to "update" their response times as the sessions went on. Figure 3D perfectly illustrates this; it shows the Response Time (RT) during the outcome generalization phase. You can see that when the environment becomes less predictable, the KO rats (the orange line/blobs) have a significantly lower RT compared to the WT rats, suggesting they are processing the information differently when the "rules" of the reward change.



The study was very well-controlled in several ways. First, the researchers used Bayesian mixed-effect models for their stats, which is a great move because it's a more rigorous way to avoid false positives. Additionally, they ensured that the people handling the rats and analyzing the data didn't know which rats were which (double-blinding), and they kept the environment strictly regulated (humidity was low, light schedules remained the same, etc.). Lastly, using the same touchscreen tasks for rats that we use for humans is another major strength. However, there are a few red flags. First, the sample size was small to begin with (20 rats), and it got even smaller after five rats were kicked out for not learning the tasks fast enough. It's hard to find subtle biological effects with so few subjects. Second, the researchers only used male rats, which means we have no idea if these findings would hold true for females. Another interesting point is that the reward used was sucrose pellets; while most rats love them, individual preferences can vary, and since sucrose activates certain receptors, it might have interacted with their 5-HTT levels in ways that weren't fully accounted for. Finally, the behavioral task was a modified version of one the researchers built themselves, which can sometimes make it harder for other labs to replicate the results exactly.

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While not everyone might care about these research findings, they should definitely be important to everybody (especially those of us in the field of neuroscience and psychology). With how prevalent and disruptive mental disorder have become in this day and age, this research provides us with two novel mechanisms for treatment. The first is that this furthers our understanding of 5-HTT's direct role in learning and environmental adaptation (updating our behaviors when expectation are not met). The second reason this research should be important to everybody is because it supports the use of generalization as yet another dimension for the growing RDoC framework. This takes us one giant step closer to reaching an age of "personalized medicine," which is important since one of the biggest downfalls of the current DSM-model is that it categorizes individuals based on symptoms and ignore their individual differences (which could actually contain the answer for treatment)

References

¹Beecher, Kate & Wang, Joshua & Jacques, Angela & Chaaya, Nicholas & Chehrehasa, Fatemeh & Belmer, Arnauld & Bartlett, Selena. (2021). Sucrose Consumption Alters Serotonin/Glutamate Co-localisation Within the Prefrontal Cortex and Hippocampus of Mice. *Frontiers in Molecular Neuroscience*. 14. 678267. 10.3389/fnmol.2021.678267.